

FastPoker

Provably Fair, Real-Time On-Chain Poker on Solana

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1. Executive Summary

FastPoker is a Texas Hold'em protocol on Solana. Every shuffle is verifiable, every payout is on-chain, every dollar of fees is paid back to the people who underwrite the venue. Gameplay runs at sub-100ms latency inside a Trusted Execution Environment, with no wallet popups for in-hand actions. The token economy is real-yield: stakers earn from real fees, never from inflation. The dealer network is permissionless: anyone can run an operator, and anyone with a license earns a share of every hand they settle.

The economic engine is a closed loop. Activity generates fees. Fees fund bonded yield. Bonded yield attracts burn-to-stake capital. That capital underwrites operators and tables. Operators and tables drive more activity. A refinement gate paces token velocity so emissions cannot front-run revenue.

Headline numbers:

- 10ms gameplay latency, gasless TEE transactions
- 5% cash rake with tiered BB caps that scale by stake size
- Stakers earn SOL, \$FP, and any approved SPL token from cash tables. Zero inflation.
- 10% refinement burn-off redistributed to patient holders
- 10% of every SNG emission funds the Royal Jackpot
- Last-out finishers earn nothing in \$FP (anti-farm baseline)
- Two Anchor programs, ~80 instructions, fully on-chain accounting

This document is the public reference for FastPoker. A companion `TECHNICAL_SPECIFICATION.md` covers byte-level account layouts, PDA seeds, instruction catalog, and error codes for integrators.

2. Problem Space

Traditional poker rooms (PokerStars, GG Poker, partypoker) gate three kinds of trust:

1. The RNG. Players cannot inspect a single shuffle. Audits exist but are produced by paid third parties on a periodic basis, not per-hand.
2. The rake destination. Where a hand's rake actually goes is set by operator policy, not verifiable on-chain.
3. The room itself. The operator is the custodian. They can freeze a stack, change rules, geo-block players, or close shop without recourse.

FastPoker addresses these by being:

- **Verifiable per hand.** Anyone can pull a hand off-chain and reconstruct the shuffle from the on-chain commitment. No periodic audit. No paid third party. The hand itself is the proof.
- **Self-custodied.** Buy-in chips sit in an escrow vault on Solana that the player can withdraw against. The room cannot freeze a stack.

- **Permissionless to operate.** Anyone can run a dealer-service binary. Anyone can hold a license and earn dealer rake. The protocol does not pick winners among operators.
- **Permissionless to verify.** Every hand report, every settlement, every rake split is public Solana state.

Prior on-chain poker projects (Virtue, ZK-Poker, CoinPoker, Decentral Games) addressed custody but were limited by the underlying L1's latency. Without a TEE, every shuffle requires a commit-reveal round-trip, which adds wallet popups and seconds of latency to each hand. FastPoker uses MagicBlock's Ephemeral Rollup TEE to keep the gameplay loop sub-100ms while preserving on-chain settlement.

ve-style real-yield protocols (Curve, Velodrome, GMX) get the yield-source story right by paying stakers from real fees. They use reversible lockups (decay curves), which lets capital cycle in and out. FastPoker uses irreversible burn-to-bond instead. Permanent infrastructure (operators, tables, the venue) deserves permanent capital.

3. Architecture Overview

FastPoker runs across two layers.

Solana L1 handles settlement, custody, emissions, and registry. Tables, deposits, cashouts, prize distribution, rake routing, dealer licenses, and auctions all settle on L1.

MagicBlock Ephemeral Rollup (TEE) handles gameplay. Tables and seats are temporarily delegated to the rollup for a hand or an SNG, then committed back to L1 for settlement. The rollup provides ~10ms confirmation latency and gasless transactions inside its scope.

Three programs do the work:

- **Anchor Poker** runs game logic. ~80 instructions covering table lifecycle, dealing, betting, settlement, jackpot triggers.
- **Steel Tokenomics** runs the \$POKER mint, staking pool, refinement, fee deposits, and reward accumulators.
- **FastPoker Registry** runs Dealer License accounts and token-listing auctions.

Two MagicBlock-side programs (Permission and Delegation) gate seat-card visibility and account state moves between L1 and the rollup.

Every table follows the same arc: created on L1, players join with a deposit, accounts delegate to the rollup, the rollup runs the hand, state commits back to L1 for settlement. SNGs reuse seats for the next match. Cash tables stay live indefinitely.

4. Provable Fairness

The trust assumption is that the TEE attestation is sound. Inside that assumption, every hand is independently verifiable.

Entropy. Each hand mixes four entropy sources: a recent slot hash, current table state, a per-hand counter, and a verifiable random function (VRF) callback from MagicBlock's ephemeral VRF oracle. The VRF program identity must sign the callback, so the entropy cannot be forged by a player or operator. The combined entropy enters the deal instruction as data visible only to the sender; opponents cannot extract it from the transaction.

Pre-deal commitment. The TEE shuffles the deck and commits a Merkle root over the result on-chain before any card is revealed.

Per-seat reveal. Each seat reads only the cards it is permissioned to read. The permission is gated by an on-chain account tied to the seat's wallet signer.

Community card reveal. Flop, turn, and river revealed in sequence as the round progresses.

Hand report. At hand end, an on-chain HandReportV1 record is posted containing the deck Merkle root, the action sequence, the pot distribution, and per-revealed-card Merkle proofs. The protocol commits a Merkle root over the full shuffled deck before any reveal, but only emits proofs for cards that were actually revealed at showdown (community cards plus shown hole cards). Mucked and folded hole cards stay private forever — they are committed to but never proven. Each table also keeps a small **Hand Ledger** account storing the rolling hash chain head, so a verifier can confirm any hand's continuity with the rest of the table's history without replaying the entire stream.

What anyone can verify, hand by hand:

1. The cards that were revealed match what the deck commitment promised, via the per-card Merkle proofs.
2. No seat read a card outside its permission scope (the on-chain permission account gates which seat can decrypt which leaf).
3. The action sequence is consistent with the rules and produces the recorded winner and pot.
4. The current hand's report chains correctly to the prior hand's rolling hash, so reports cannot be silently rewritten.

What anyone cannot verify (by design): the actual cards held by a player who folded or mucked. Those cards are committed but never opened. This matches normal poker privacy: opponents do not see your folded cards.

What fairness covers: shuffle integrity, card visibility, deal/reveal ordering, action sequence consistency.

What fairness does not cover: human collusion (two players sharing hole cards via an external channel is undetectable on-chain), bot detection (the protocol does not classify play styles), and TEE compromise (a compromised validator could in theory equivocate; the Merkle commitment makes equivocation detectable post-hoc but cannot prevent it in-stream).

5. Cash Game Lifecycle

Cash tables are the high-volume revenue surface. They follow a different lifecycle than tournaments.

A cash table is created once and stays live indefinitely. Players sit down by depositing chips into an escrow vault on L1, then their seat is delegated to the rollup for gameplay. Players can sit out, sit back in, top up a short stack, or leave and claim remaining chips back to their wallet via a pull-claim flow. The table itself never resets; only the seats turn over.

Cash tables can use SOL, \$FP, or any approved SPL token. The protocol earns rake denominated in the asset being played. Every approved-token cash table feeds the same staker reward pipeline: rake routes through the Steel program, which distributes per-mint reward accumulators, and bonded stakers can claim their share in any mint.

Stack management is BB-relative for display but lamport-exact on-chain. Big blinds escalate inside SNGs; cash games hold the blind structure constant for the lifetime of the table.

The cash settle cycle runs at every hand: rake is computed, distributed to the four recipients (creator, dealers, stakers, treasury), and the next hand begins. Settlement is atomic. One L1 transaction routes the entire cash flow per hand.

User-created tables. Anyone can create a cash table via `create_user_table`, paying the creation fee (Section 10). The creator becomes the rake recipient for the 50% creator share on that table for its lifetime. User-created tables can be **public** (anyone may sit) or **private** (whitelist-only). Private tables enforce membership at join time: an on-chain whitelist account gates `join_table`, and only wallets the creator has explicitly added (creator implicitly whitelisted) can sit down. The creator can manage the whitelist post-creation via `manage_whitelist`. Private tables let players run home games, league nights, or invite-only stakes while inheriting the same rake split, fairness, and settlement guarantees as public tables.

Tiered Rake Cap

Cash rake is 5% of the pot, capped at a tier-dependent BB multiple. The tier is selected by the table's big blind size:

Tier	BB range (SOL)	HU cap	6-Max cap	9-Max cap
Micro	$BB \leq 0.01$	uncapped	uncapped	uncapped
Low	$BB \leq 0.025$	3 BB	5 BB	8 BB
Mid-Low	$BB \leq 0.05$	2 BB	4 BB	6 BB
Mid	$BB \leq 0.1$	1 BB	2.5 BB	4 BB
Mid-High	$BB \leq 0.5$	0.5 BB	1 BB	1.5 BB
High	$BB \leq 1$	0.25 BB	0.5 BB	0.75 BB
Nosebleed	$BB > 1$	0.09 BB	0.15 BB	0.20 BB

Caps shrink in BB terms as stakes grow. At Micro stakes, the room takes the full 5% with no ceiling because absolute amounts are small. At Nosebleed stakes, the cap is fractions of a BB because absolute amounts are large and players are sensitive to per-hand cost.

SPL tables fall into two buckets. Tokens explicitly approved as "premium" by governance (currently SOL and \$POKER) get parallel tier tables with their own boundaries and caps. Tokens that won the right to a cash table via the listing auction system (Section 13) use a flat 5% uncapped rake — they pay the full percentage on every hand. The distinction lets the protocol concentrate cap protection on its core assets while extracting full economics from the long tail of listed pairs.

6. SNG Pools and Tournament Play

Sit-and-Goes (SNGs) work differently from cash. Players don't pick a specific table; they join a pool for a format and tier. When the pool fills, a match triggers and N players are seated together at a fresh table.

Three properties matter:

Random seat assignment. When a pool fills, the protocol runs a Fisher-Yates shuffle over the queued players to determine seat order. The shuffle is deterministic from the on-chain trigger entropy, so anyone can verify the seat assignment was not gamed.

Gasless first-hand action. SNG players want to act on hand 1 without a wallet popup. Each player's pre-approved session signer travels through the pool's matching pipeline and is written into the seat at match time, alongside the player's wallet. When the match triggers, every seat already has its session signer set; the player can fold, call, or raise on the first hand without signing a transaction.

Table reuse. When an SNG ends, the table doesn't get destroyed. The seat layout returns to L1, the deck and card state stay cached on the rollup, and a fresh pool fill bootstraps the next round on the same physical table account. This dramatically reduces account creation overhead at scale.

Tier pricing across formats follows a fixed ladder (Micro, Bronze, Silver, Gold, Platinum, Diamond) so liquidity concentrates instead of fragmenting across arbitrary buy-ins.

7. MagicBlock TEE: Real-Time Gameplay

The MagicBlock Ephemeral Rollup is what makes sub-100ms poker possible on a public chain.

When a hand starts, the relevant accounts (Table, Seat, SeatCards, DeckState, SlimBuffer) are delegated to the rollup. The TEE owns them for the duration of gameplay. Transactions inside the rollup confirm in approximately 10ms and cost nothing in gas. When the hand or SNG ends, accounts undelegate back to L1 and the final state commits.

A few specifics shape protocol design:

- The frontend subscribes on `processed` commitment, not `confirmed`. Inside the rollup, `processed` fires sub-100ms; `confirmed` adds latency without adding value because the rollup's reorganization risk is structurally different from L1.
- Account delegation order matters. Sub-accounts (SlimBuffer, Permissions, Seat, SeatCards, DeckState) follow a defined ordering before the parent table delegates, so all writes have a consistent state to land into.
- Cards and decks stay rollup-resident across SNG matches. When an SNG resets for the next match, only the table-level seat layout round-trips through L1. The actual card state stays on the rollup.

The trust root is the TEE attestation. Validator decentralization is on the MagicBlock roadmap and is the largest structural risk for the protocol until then.

8. Token Design (\$FP)

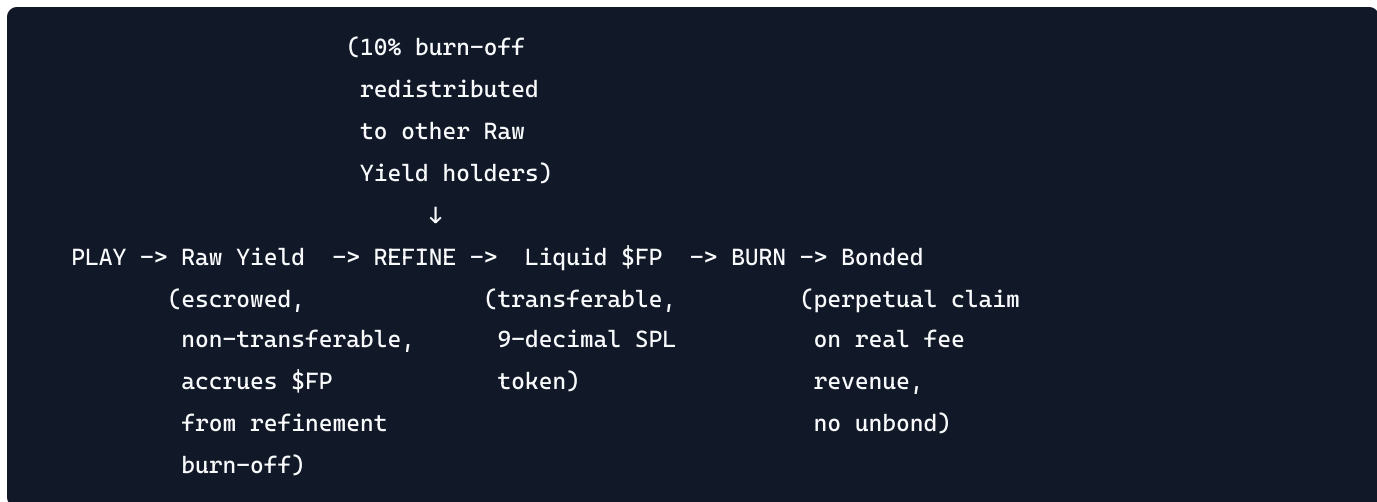
Mint Properties

- Symbol: \$FP
- On-chain mint: POKER, 9 decimals
- Mint authority: a program-derived address, no human key
- Initial supply: 0 (no pre-mine)
- Issuance source: SNG play only. Cash games never mint.

- Hard cap: none. Supply is shaped by the decay curve and asymptotically bounded.

Three-State Model

\$FP exists in three distinct states.



State	Description
Raw Yield	The earnings from \$FP-emitting play, held in pre-refinement form on the player's account. Cannot be transferred or burned. While held, accrues a share of other refiners' burn-off as liquid \$FP.
Liquid \$FP	Post-refinement form. Standard SPL token. Transferable, swappable, burnable.
Bonded	Burned-to-stake position. Permanent. Holder accrues a perpetual claim on real protocol revenue streams.

The Refinement Gate (the Dampener)

When a player refines Raw Yield into Liquid \$FP, the contract takes a 10% burn-off and redistributes it pro-rata to all remaining Raw Yield holders. This single mechanism does three things:

1. Paces token velocity. New emissions cannot become liquid the moment they are earned. The supply curve trails the fee curve instead of front-running it.
2. Rewards patience. Holders of Raw Yield collect dust from impatient refiners. Patient holders harvest the impatient.
3. Aligns supply with real-yield demand. The demand to bond pulls supply through the gate, since liquid \$FP is what you bond to earn real yield.

This is the central novel mechanic of the protocol. Other on-chain poker games have inflation, vesting, lockups, or buybacks. None of them have a dampener that pays its tax to the most patient participants.

Permanent Bond

Burning \$FP to stake is irreversible. There is no unbond, no decay, no scheduled unlock. Three reasons:

1. Permanent infrastructure deserves permanent capital.
2. Reversible stake invites mercenary cycling. Capital arrives during high yield, leaves during dry spells, leaving the protocol underwritten by no one in the gap.

3. Without a scheduled unbond cliff, there is no liquidity event for stakers to coordinate around. Today's bonded supply is locked in forever.

9. Emission Schedule

\$FP issuance is gated by SNG play. Cash play generates rake (a fee, not an emission) and never mints new tokens.

Per-seat base emission rates (in 6-decimal micro-units of Raw Yield):

Format	Players	Per-seat base
Heads-Up	2	40
6-Max	6	120
9-Max	9	180

Total table emission = `per_seat_base × max_players`.

The base is multiplied by a decay function:

```
mult = (1 - circulating / 1B) ^ 1.2
```

Floored at 5% of base. The decay reads circulating supply live at settlement time, so emissions taper as supply matures.

10% of every SNG's gross emission is skimmed before player distribution and added to the global Royal Jackpot pool. The remaining 90% is the player pool.

Two design choices to highlight:

- **No buy-in scaling.** A 0.01 SOL Heads-Up SNG mints the same \$FP as a 1 SOL Heads-Up SNG. Emission is per-seat, not per-dollar. Whales cannot buy disproportionate token issuance.
- **Cash games never mint.** Cash play is unbounded; tournament play is bounded. Limiting issuance to tournaments ensures each minted unit corresponds to a discrete competitive event.

10. Revenue Architecture

The protocol has four real fee streams. Every one of them pays bonded stakers a meaningful share. Nothing is funded by inflation.

Cash Game Rake

5% of every cash pot, with tiered BB caps as described in Section 5. Split:

Recipient	Share
Table creator	50%
Dealers (operators who cranked the hand)	20%
Bonded stakers	20%
Treasury	10%

For SOL tables, the staker share lands in the Steel pool's SOL reward accumulator. For approved SPL tables, the staker share lands in a per-mint SPL reward accumulator. Stakers claim each currency separately on a pull basis.

SNG Entry Fees

Recipient	Share
Bonded stakers	45%
Dealers	45%
Treasury	10%

Largest single staker stream by volume. Paid in SOL.

Dealer License Sales

Each license sale routes to the Steel program's revenue distributor, which splits 50/50:

Recipient	Share
Bonded stakers	50%
Treasury	50%

Edge case: if no stakers exist yet, the full sale routes to treasury. Once any wallet has bonded, the split becomes 50/50.

Token-Listing Auction Fees

Same 50/50 split:

Recipient	Share
Bonded stakers	50%
Treasury	50%

Atomic Settlement

All cash game rake routing happens inside a single atomic L1 transaction. Operator weights are read from on-chain rake-tally accounts. Multi-operator settlements use chunked sub-transactions to stay under Solana's per-transaction size limits.

Other Surfaces

Two additional money flows exist that aren't part of the four protocol revenue streams above:

Table creation fee. A user creating a custom cash table pays a flat 0.05 SOL via the Steel program (which routes 50/50 to stakers and treasury per the standard split) plus a small denomination fee (1-2 BB in the table's token) that helps seed the vault. This fee discourages spam table creation and gives stakers a cut of the venue expansion. System tables (created by the protocol) pay no fee.

TipJar. Each cash table has an optional TipJar account funded by players who want to tip the operator that's cranking their table. TipJar deposits flow directly to the licensed dealer who cranks each hand, paid per-hand from the jar. This is a player-to-dealer incentive surface, not protocol revenue. SOL only, cash games only, and tips refund to the table creator if the table closes with leftover balance.

11. SNG Prize Curves

After the 10% Royal Jackpot skim, the remaining 90% of each SNG's emission is the player pool, distributed via two parallel structures.

SOL Prize Splits

The buy-in pool, paid only to ITM (in-the-money) finishers:

Format	1st	2nd	3rd
Heads-Up	100%	—	—
6-Max	65%	35%	—
9-Max	50%	30%	20%

\$FP Emission Splits

The Raw Yield pool, paid to a wider trickle of finishers:

Place	HU	6-Max	9-Max
1st	100%	55%	50%
2nd	0%	25%	25%
3rd	—	12%	15%
4th	—	5%	5%
5th	—	3%	3%
6th	—	0%	1%
7th	—	—	1%
8th	—	—	0%
9th	—	—	0%

Every paid place earns Raw Yield. The hard zeros on bottom seats prevent register-and-fold farming. A bot that busts on hand 1 of a 9-max lands in 9th and earns nothing. Players who outlast at least one bust earn a meaningful trickle, even outside the SOL ITM band.

12. Dealer Licenses and the Operator Economy

The protocol is permissionless to run, but only licensed dealers earn dealer rake.

A dealer's job is to watch L1 and TEE state and submit the right instruction at the right time: deal a card, settle a hand, distribute prizes, trigger a jackpot, clear a timeout. Cranks are automatable. Most dealers run a self-hosted dealer-service binary the protocol maintains.

Licenses are minted via the FastPoker Registry program. Pricing follows a linear bonding curve: each new license costs a fixed increment more than the previous one, capped at a hard maximum. Current parameters:

Parameter	Value
Base price	0.001 SOL
Increment per license	0.001 SOL
Max price	~9.9 SOL

Formula: `price = base + total_sold × increment`, clamped to max. The full sale price is split 50/50 between bonded stakers and treasury, so launching a new license actively rewards early participants.

In v1, licenses are stored as Anchor-typed accounts in the registry program, not as tradable NFTs, and they are non-transferable. This is intentional. After the license sale fully concludes, a mintable NFT layer may be added to enable secondary-market transfer. Until then, license ownership is bound to the wallet that purchased it.

A licensed dealer earns from two streams:

Stream	Share
Cash rake	20% (weighted by hands cranked at each table)
SNG entry fees	45% (same weighting)

Earnings accrue to per-operator on-chain accounts. Dealers withdraw via a pull-claim transaction on their own schedule. The dealer-service dashboard provides earnings visualization, license status, and table list locally. Nothing about a dealer's economics is mediated by a central server.

Operator selection at any given table is weighted by hands cranked. A dealer who cranks the most hands at a table earns the largest share of that table's dealer pool. This rewards uptime and responsiveness, not mere license ownership.

13. Token-Listing Auctions

The FastPoker Registry program runs auctions for SPL-token cash table listings. Each round runs for a configurable duration; bidders deposit toward a target mint they want listed, and at the end of the round the protocol promotes the top-bid mints into the active listing slate. Three properties shape the design.

Bid deposits route immediately. When a wallet bids on a mint, the bid amount routes through the Steel program's revenue distributor in the same transaction. Stakers and treasury get their cut of the bid the moment it lands, regardless of whether that mint wins the slot.

Carryover and ranking. Active bids accumulate per mint over the course of a round. At round close, mints are ranked by total bid; the top N enter the active listing set, displacing whichever incumbents fell below the threshold. Wallets that bid on losing mints can carry their stake forward into the next round rather than rebidding from zero.

Already-listed mints cannot be re-listed. If a mint is currently active, the program rejects bids on it. The active set turns over only through ranking changes between rounds, not through one-shot challenges.

Adaptive duration decay. Round duration self-tunes based on bid intensity. If a round closes with higher total bid than the previous round, the next round shortens by a configured step value. If lower, the next round lengthens. Bounded by configured min and max durations. The protocol speeds up when demand is hot and slows down when demand cools.

Listing fees split 50/50 between bonded stakers and treasury (the fourth real revenue stream into the staker pool). The mint must conform to standard SPL invariants (no rebasing, no transfer hooks unless explicitly allowlisted).

The mechanism serves two purposes: it bootstraps liquidity for permissionlessly-listed SPL pairs (which pay the flat uncapped 5% rake described in Section 5), and it creates a recurring treasury revenue stream independent of poker volume.

14. Jackpots

Two jackpots run concurrently. Both are funded by SNG activity; cash games do not contribute.

Lucky Jackpot (SOL)

Funded by per-SNG opt-in contributions. Triggers hand-by-hand inside settlement, with deterministic on-chain odds. Eligibility requires the seat to have opted in and to be in the active hand mask at the moment the trigger fires (still in the hand, not folded). When triggered, the pool splits equally among eligible seats and credits each player's claimable SOL balance.

Royal Jackpot (Raw Yield)

Funded by 10% of every SNG's gross emission. The pool is denominated in Raw Yield (unrefined POKER, 6-decimal micro-units), not liquid \$FP. When a winner claims, they receive Raw Yield and can refine to liquid \$FP through the standard 10% refinement gate. Triggers via a probabilistic roll inside settlement. Eligibility includes every seat that was active during the trigger hand, including seats that bust later in the SNG. The acc-per-weight accumulator credits each eligible seat's share before the seat's weight is removed from the active pool at SNG end.

The Royal mechanism is the most player-favorable jackpot model in on-chain gaming. A losing player can still walk out of an SNG with a Royal share if the trigger fired during their participation, regardless of where they finished. Royal is "you were in the room when it happened," not "you won the hand."

15. Bonded Staker Yield (the Demand-Side Anchor)

This is the layer that makes the tokenomics non-circular.

A bonded position earns a perpetual claim on four real-revenue streams:

Stream	Source	Currencies
Cash rake	20% of cash pot rake across all approved cash tables	SOL, \$FP, approved SPL tokens
SNG entry fees	45% of tournament fees	SOL
License sales	50% of dealer license revenue	SOL
Auction fees	50% of token-listing fees	SOL

Plus a \$FP component from the refinement burn-off, which redistributes to Raw Yield holders (some of whom are also bonded).

No inflation. The protocol does not mint \$FP to pay stakers. Yield is sourced entirely from fees paid by active players.

Why Permanent Bond

The bond is irreversible by design.

1. Permanent infrastructure deserves permanent capital.
2. Reversible stake invites mercenary cycling.
3. Without an unbond cliff, there is no liquidity event for stakers to coordinate around. Today's bonded supply is locked in forever.

How Yield Accrues

The protocol uses an accumulator-per-weight pattern. Each fee deposit increments a global accumulator by $\text{deposit} / \text{total_bonded_weight}$. Each staker's pending claim is $\text{bonded_weight} \times (\text{current_acc} - \text{snapshot_at_bond})$. Constant-time deposits, constant-time claims. This is the same pattern Curve, Velodrome, and GMX use for the same problem. It is gas-efficient and battle-tested.

Pull-Claim

Earnings live on the staker's account, not in a withdrawable global balance. To realize earnings, the staker signs a claim instruction with their wallet on L1. Claims pay normal Solana transaction fees (a few thousand lamports) and are not session-key gasless. Session-key gaslessness applies only to in-hand TEE actions (fold, check, call, raise, post blind, etc.); L1 operations like joins, deposits, cashouts, claims, staking, and auctions all require a wallet signature and pay L1 gas.

16. Off-Chain Stack

The on-chain code is the source of truth. Off-chain components exist to make the on-chain truth usable.

Crank service. A long-running process that watches L1 and TEE state and submits crank instructions: timeouts, settlements, jackpot triggers, prize distribution. It exposes no public HTTP or WebSocket surface.

The contract enforces that the crank cannot lie. Every instruction validates state from on-chain accounts. Each licensed dealer runs a variant called dealer-service.

Indexer. A Mongo-backed service listens to the program's events and writes them to collections. Events tracked include hand settled, prizes distributed, rake distributed, player joined/left, jackpot triggered, table created/closed. The indexer powers leaderboards, profile career stats, dealer health, and jackpot history.

Discord integration. A bot links Discord identities to wallets via signed challenge, syncs roles based on player level and tier, runs a find-game wizard, posts jackpot alerts and dealer health snapshots, and renders leaderboards across SOL and \$FP categories.

Frontend. A Next.js client renders the lobby, table, profile, earn, leaderboard, jackpots, dealer info, admin, my-tables (private table management), how-to-play, and verify pages. Game state uses a hybrid pattern: WebSocket subscriptions for sub-100ms reactivity, with L1 polling as a backstop. The client never trusts a backend for game state; everything is read from L1 or TEE directly.

The product surface around the on-chain core is wider than just the table:

- **Identity layer** (profile, avatars). Each wallet can claim a username, pick from generated/curated/NFT avatars (NFT avatars unlock at level 5+, with a community-submitted allowlist of approved collections). Identity persists across sessions, syncs to leaderboards, and shows up at the table.
- **Career stats** (hands played, hands won, sessions, lifetime SOL profit, \$FP earned, ITM count, tournaments won, ROI, achievements). Sourced from the indexer.
- **XP and tier system.** Players earn XP from cash hands and SNG finishes; XP unlocks levels and a tier ladder (fish/reg/shark/pro/ /...). Tier syncs to Discord roles for community recognition.
- **Chat service.** A standalone chat server with wallet-signature authentication and a Mongo audit log. Per-table chat plus protocol-wide channels. Per-IP rate limit and history hydration on rejoin.
- **Admin and dealer dashboards.** Internal admin view for monitoring tables and config. Public `/dealer` info page with live network health stats. Each licensed operator runs their own self-hosted dealer-service dashboard locally.
- **Hand history, hand replay, fairness verifier.** Any hand can be pulled from chain and replayed with action-by-action visualization on the verify page.
- **My-tables / private table management.** Creators of user-tables manage whitelists, monitor table health, top up TipJars, and close tables when finished.

17. Risks and Limitations

Trust Assumptions

The fairness guarantee relies on the MagicBlock TEE's attestation. A compromised validator could in theory equivocate. The Merkle commitment makes it detectable post-hoc but cannot prevent it in-stream. Validator decentralization is on the MagicBlock roadmap and is a structural risk for the protocol until then.

If the TEE goes offline, gameplay halts. Settlement on L1 is unaffected, and players can recover deposits via on-chain recovery instructions. Live tables become unavailable until the rollup recovers.

Smart Contract Risk

Internal security review was completed in April 2026. Identified vulnerabilities (rake-route ATA owner check, SngPool destruction guard, deposit delegation guard) were patched and re-deployed. A bug bounty program is planned but not yet active.

Upgrade authority on the Anchor poker, Steel, and Registry programs is held by a Squads V4 multisig (3-of-5, tiered timelock) as of 2026-04-24. The deployer keypair no longer has direct admin authority. All admin instructions route through Squads proposals.

Economic Attack Vectors

- **Refinement front-running.** A whale could pre-load Raw Yield, refine in size, and capture the bulk of one burn-off cycle. The mitigation is structural: burn-off pays current Raw Yield holders, not by holding period. Sustained holders accumulate from many small refines, not one large one. Front-running pays itself.
- **Operator concentration.** With a small dealer set, one operator can capture a disproportionate share of cranks. The bonding-curve license pricing makes maintaining a monopoly economically expensive. The per-table rotation prevents any single license from dominating.
- **Sybil cranking.** A single entity running multiple licenses is allowed. The bonding curve makes it expensive. The hand-weighted rotation prevents any single license from dominating.
- **Refinement burn perception.** Holders may view the 10% burn-off as a tax. The narrative needs to be clear: the tax is paid TO Raw Yield holders, not into a void. Patient holders are the recipients.

Regulatory

- **Skill vs chance.** Texas Hold'em is a skill game in some jurisdictions, gambling in others. Jurisdictional gating is the operator's responsibility and is enforced at the dealer-service level on a per-operator basis.
- **\$FP utility framing.** \$FP holders earn from real fees, not from token speculation. The framing matters for securities classification: the token is utility (refinable, bondable, claim on revenue), not a passive yield instrument. Classification depends on jurisdiction.
- **KYC.** The protocol does not enforce KYC. Operators may; specific tournament series may; the website may.

Mainnet Readiness Gaps (current)

- **Operator reward audit.** The unified `CrankAction` reward path (split L1/ER variants reconciled by `finalize_operator_reward_epoch`) replaced an earlier patched design in May 2026 and supersedes any earlier "Option A / Option B" framing. The new design needs a final audit pass against weight dilution, ER replay, and caller-to-recipient bind before mainnet.
- **Secrets cleanup.** Pre-public-launch rotation and history purge of any deployer-era secrets still pending.
- **SNG license-path validation.** Licensed-dealer SNG settlement code is complete but has not yet been exercised end-to-end on devnet (no live licenses on devnet at the time of writing).

The Squads V4 multisig migration is complete (post-2026-04-24). The multi-operator on-chain tally was validated in production via per-operator PDAs.

Honest Comparison

Dimension	Traditional rooms	Prior on-chain poker	ve-protocols	FastPoker
Provable RNG per hand	No	Partial (commit-reveal)	N/A	Yes (TEE + VRF)
Custody	Centralized	Mixed	Self-custody	Self-custody
Latency	Low	Multi-second	N/A	Sub-100ms (TEE)
Yield source	N/A	Inflation or none	Inflation or fees	Real fees only
Stake reversibility	N/A	Varies	Reversible (decay)	Permanent burn
Rake transparency	Opaque	Variable	N/A	Fully on-chain
Operator permissionless	No	Mixed	N/A	Yes

18. Roadmap and Launch

Mainnet Gating Checklist

1. Pull-claim Option B (replay-protected staker withdrawals)
2. Secrets rotation and git history purge before public launch
3. End-to-end licensed-dealer SNG path validated on devnet
4. Final security review pass on changes since April 2026
5. Bug bounty live before mainnet program ID rotation

Phased Mainnet Rollout

The launch sequence prioritizes SNGs and the dealer economy.

- **Phase 1.** SNGs go live alongside the initial dealer license drop. Token issuance starts. Refinement and Royal Jackpot are active. Dealers earn SNG entry fees from the first match.
- **Phase 2.** Cash games unlock for SOL and \$FP. Cash rake begins flowing to creators, dealers, stakers, and treasury. Cash table creation may be gated initially (allowlisted creators only) and opened progressively.
- **Phase 3.** Token-listing auctions unlock. Approved SPL cash tables come online. Multi-asset rake diversifies treasury revenue.
- **Phase 4.** Open dealer license auction continues to scale the operator network. Secondary-market license transfer (if shipped) lights up here.

The phasing is deliberate. Issuance and the operator economy boot together (so the protocol has both supply and infrastructure from day one). Cash and SPL diversification follow once the SNG flywheel is producing real revenue. Open license trading comes last, after the initial license sale has run its course.

Bootstrapping

- **Initial dealer set.** A small set of seeded operators run cranks during early phases to ensure liveness.
- **Treasury runway.** SOL reserves underwrite operator gas, ATA creation, and audit costs during early-revenue phases.
- **No pre-mine.** \$FP supply is 0 at mainnet genesis. All issuance comes from Phase 1 SNG play.

Post-Mainnet

Pot-limit Omaha and other variants. Multi-table tournaments. Mobile app. Mainnet KYC partner integration for jurisdiction-specific compliance. Governance, deferred until protocol revenue is structurally significant.

Companion Documents

- `TECHNICAL_SPECIFICATION.md` — byte-level account layouts, PDA seeds, instruction catalog, error codes, event schemas
 - `ARCHITECTURE.md` — deeper engineering walkthrough of the L1/ER state machine
 - `ECONOMICS.md` — deeper economic modeling, scenario analysis, parameter rationale
 - `SNG_FP_EMISSIONS_PLAN.md` — design rationale for the non-ITM trickle table
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